

Sumedh Kothari

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EDUCATION

Georgia Institute of Technology

B.S. in College of Engineering

Expected May 2028

GPA: 3.82

- **Coursework:** Computer Organization and Programming; Computing for Engineers; Advanced Data Structures, Embedded Systems, and Networking; Ordinary Differential Equations
- **Honors:** Symposium of Rising Scholars Presenter, Eagle Scout

EXPERIENCE

Georgia Tech Renewable Bioproducts Institute | Tong Laboratory

Undergraduate Research Assistant

Atlanta, GA

August 2025 – Present

- Used Python, NumPy/Pandas libraries to apply Bayesian statistical modeling to optimize lignin valorization reaction
- Utilized NMR, SEM, and HPLC modeling data to characterize value-added chemicals from lignin breakdown
- Completed comprehensive literature reviews on aromatic compound synthesis, 3D printing custom GO structures, and optimal catalyst design while collaborating with research team to write manuscript for publication

HyTech Racing | Georgia Tech

Embedded Systems Engineer

Atlanta, GA

February 2026 - Present

- Designed and implemented a real-time lap tracking module in modern C++ for Georgia Tech's Formula SAE Racing team, leveraging Protobuf for low-latency telemetry serialization and inter-process communication across vehicle subsystems
- Developing and deploying a YOLO-based object detection pipeline for driverless perception, enabling real-time cone detection and localization through C++ inference integration into the autonomous control stack

ChemE Cube | Georgia Tech

Team Lead – Mechanical

Atlanta, GA

August 2025 – Present

- Leading the mechanical development of a 1 ft³ direct air capture reactor by managing a 5-member subteam and overseeing end-to-end CAD, prototyping, and systems integration, delivering a fully fabricated and test-ready prototype
- Designing airflow distribution and sorbent containment subsystems using parametric CAD, 3D printing, and CNC machining, achieving 75% reduction of ambient CO₂ in bench-scale validation for AIChE competition (Nov 2026)

PROJECTS

GameSense - Winner - TechHack Valley 2026 Hackathon - Hardware Track | Georgia Tech

Mechanical/Software Engineer

Atlanta, GA

March 2026

- Developed GameSense, a real-time accessibility interface for deaf / hard-of-hearing gamers that converts spatial game audio → multi-channel haptic feedback, enabling directional sound perception through vibration
- Engineered a full wearable hardware stack in 36 hrs: CAD-modeled, 3D printed a custom headband + electronics enclosure (Fusion), integrated ESP32 microcontroller, 9-axis IMU, and 8-motor vibration array, and wrote C++ firmware for real-time sensor acquisition, motor PWM control, and UART serial communication with the host audio-processing pipeline
- Developed real-time signal processing and control algorithms: implemented multi-channel audio detection and amplitude → haptic intensity mapping, programmed head-tilt → WASD movement using a Kalman filter to stabilize motion input and reduce sensor noise

IEEE Robotech 2026 Hackathon - Autonomous Track | Georgia Tech

Mechanical/Software Engineer

Atlanta, GA

January 2026

- Designed and CAD-modeled a fully autonomous debris-clearing robot by engineering the chassis, actuation mechanisms, and beacon-deployment system, enabling end-to-end autonomous operation without human intervention
- Trained, developed, and deployed a YOLO-based computer vision model in PyTorch by integrating it into the onboard perception pipeline, achieving 95% debris detection accuracy and enabling real-time hazard identifications
- Integrated perception, navigation, and mechanical actuation systems by linking CV outputs to motion planning and debris-removal hardware, allowing the robot to autonomously clear obstacles to certify landing safety

EcoFlow - Sustainability Calculator | Python, FastAPI, Next.js, Redis, PostgreSQL

Developer

Atlanta, GA

January 2026

- Built and deployed a FastAPI (Python) + Next.js (TypeScript) full-stack web app that processes any Amazon ASIN across 5 lifecycle stages, automating cradle-to-grave LCA calculations and reducing analysis time from hours to seconds via AI-driven product parsing and structured emissions modeling
- Engineered asynchronous REST APIs with Redis caching (24-hr TTL) and PostgreSQL persistence, containerizing services with Docker to achieve <1s repeat-query latency while improving scalability and cost efficiency

AiApply - AI Job Applier | Python, Selenium WebDriver, HTML, JavaScript

Developer

Atlanta, GA

October 2025

- Built full-stack AI platform automating internship applications: resume optimization, cover letter generation, Handshake submission
- Engineered Selenium-based web scraping pipelines to extract recruiter profiles and identify target companies by location/industry, then automated personalized direct messages and tailored email outreach at scale with candidate skillsets dynamically parsed from resume

Sports Betting Arbitrage Calculator | Python, REST APIs

Developer

Atlanta, GA

November 2025

- Built application aggregating live sportsbook odds via API to identify risk-free arbitrage opportunities with up to 50% daily ROI
- Engineered production-ready pipeline with Docker, session management, and logging for scalable real-time sports markets analytics

SKILLS

- **Technical:** Python, C++, FastAPI, Next.js, Redis, PostgreSQL, Data analysis (Excel, MATLAB), CAD (SolidWorks, Fusion)
- **Communication:** Technical Writing, Multilingual (English, Hindi, Spanish)